

Kubernetes 社区新动向

- AI Gateway 与 Conformance 工作组成立
- DRA GA

徐俊杰

Open Source Team
Lead, DaoCloud

2025/11/15

Patrick Ohly

Cloud Software
Architect, Intel





Patrick Ohly



Pre Steering Committee Member
SIG Testing Tech Lead
WG Device Management Chair
DRA Core Maintainer



Paco Xu



Steering Committee Member
Kubeadm Maintainer
SIG Node Reviewer
CNCF TAG Workloads Foundation Chair
LFAPAC Evangelist

The Steering Committee



Antonio Ojea
[@aojea](#)

Google



Benjamin Elder
[@BenTheElder](#)

Google



Rita Zhang
[@ritazh](#)

Microsoft



Maciej Szulik
[@soltys](#)

Defense Unicorns



Paco Xu
[@pacoxu](#)

DaoCloud



Sascha Grunert
[@saschagrunert](#)

Red Hat



Kat Cosgrove
[@katcosgrove](#)

Minimus

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01 **Kubernetes** 社区简介

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Community Snapshot - Nov 2025

97.8K

Contributors

4.63M

Contributions

8.6K

Reviewers

Stats from [Kubernetes Devstats](#), Nov 10 2025

Community Structure



- The Foundation which Kubernetes is part of.
- Non-Profit Organization
- Provides/manages funding & various guidance & services

Steering Committee

- 7 Members
- Elected by Kubernetes Org Members
- Interfaces with CNCF
- Whole-Project Scope

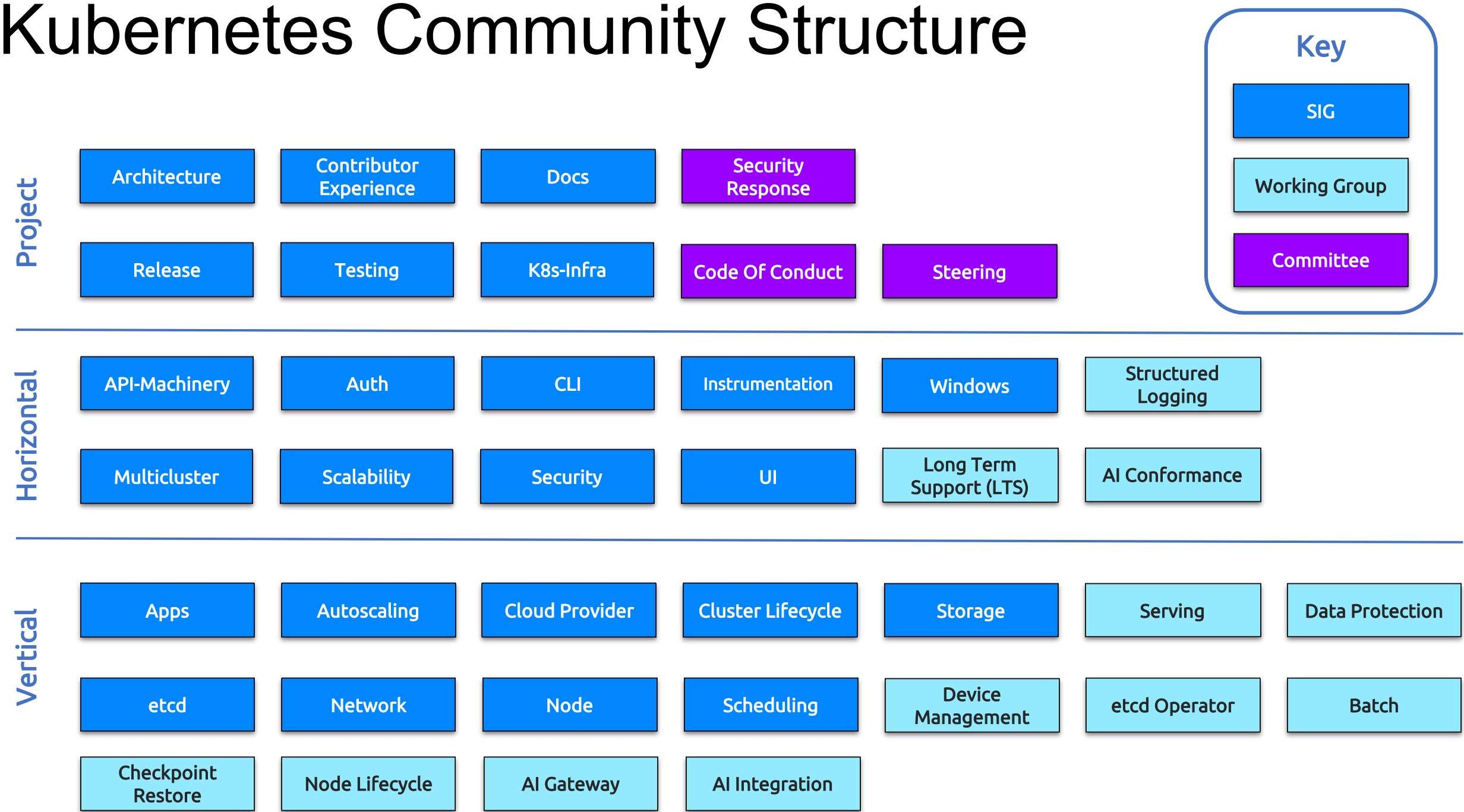
SIGs / WGs Special Interest / Working Groups

- Work Areas within Kubernetes
- IE. Networking, Release, Docs, etc.
- SIGs = constant
- WGs = topical/temporary

Subprojects

- Created by SIGs.
- Groups that execute on specific work/projects.

Kubernetes Community Structure



In Memoriam



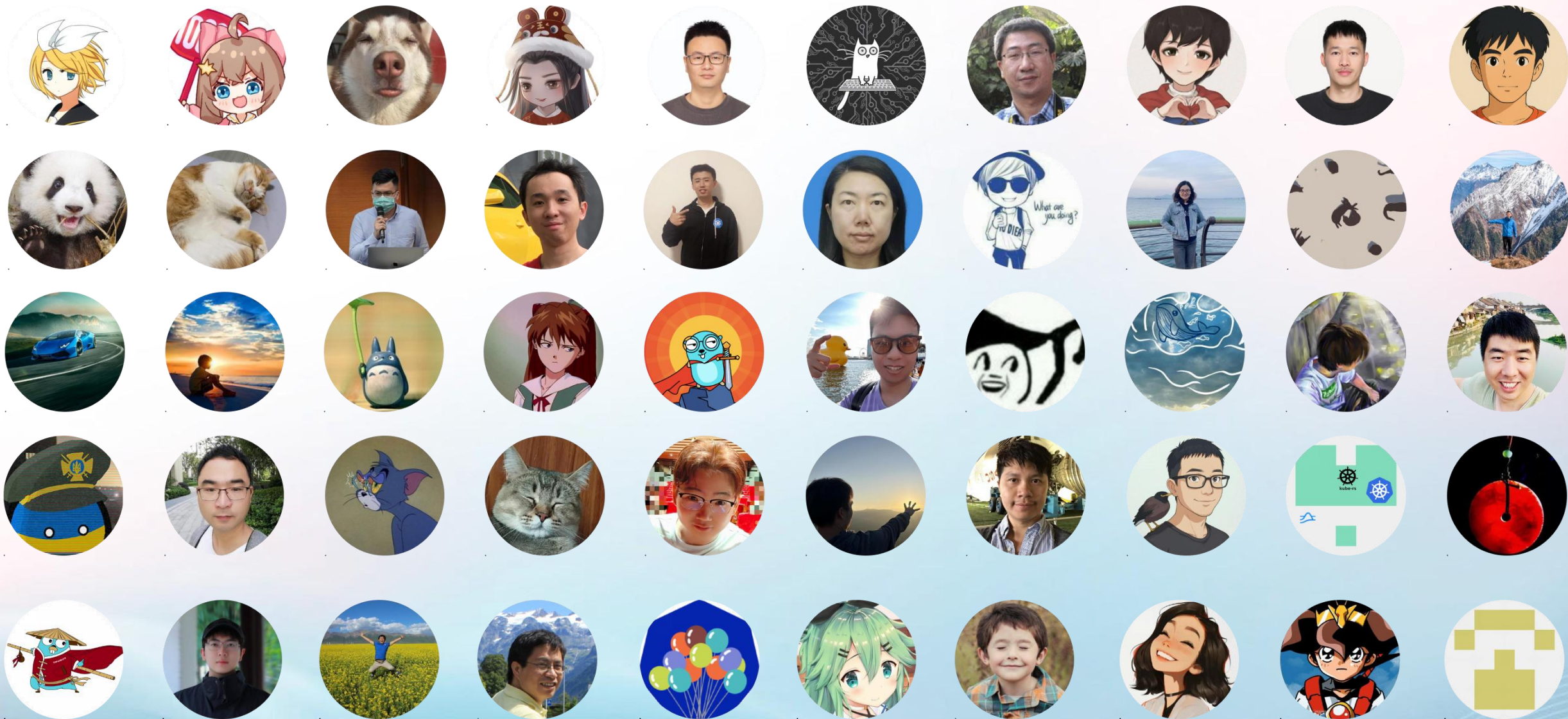
Han Samuel Kang



Rodolfo Martínez Vega



感谢中国贡献者的贡献



Kubernetes Contributor Award 2025

[@HirazawaUi](#)

Bing Hongtao

SIG Cluster Lifecycle

HirazawaUi has excelled at being a new kubeadm contributor for SIG Cluster Lifecycle. HirazawaUi climbed the contributor ladder of being first a reviewer and then an approver very fast, while actively reviewing new PRs and participating in important discussions.



Bing Hongtao

SIG Node

In recognition of your impactful work on critical bug fixes, development of new features, and dedicated efforts in testing evented PLEG to assess its viability.

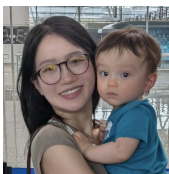


[@sunnylovestiramisu](#)

Sunny Song

SIG Storage

Sunny Song (github: sunnylovestiramisu slack: Sunny Song) has been driving the design and implementation of the VolumeAttributesClass feature, a critical enhancement slated for GA in Kubernetes 1.34. This involves collaboration with multiple cloud providers and storage vendors. Sunny was instrumental in bringing the volume populators feature to GA in the Kubernetes 1.33 release, further solidifying the platform's capabilities in managing dynamic storage needs. She also provides ongoing support for Kubernetes CSI sidecar releases and resolves various bug fixes.

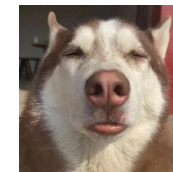


[@googs1025](#)

Jiang Zhenyu `CY Jiang`

SIG Scheduling

For making tons of contributions in Kubernetes scheduler, and also being active in other sub projects as well, such as descheduler and scheduler-plugins. Thanks for all the contribution!



[@yongruilin](#)

Yongrui Lin

SIG API Machinery

We nominate Yongrui Lin for his significant and detailed contributions to the implementation of declarative validation in Kubernetes. Working closely on KEP-5073, Yongrui has been pivotal in the hands-on development and refinement of the validation-gen tool and its integration into the Kubernetes codebase. His numerous commits over the past 12 months reflect a deep commitment to improving the API validation process. Yongrui's technical diligence and collaborative efforts have been crucial in turning the vision of a declarative validation system into a tangible reality, directly impacting the future maintainability and stability of the Kubernetes API.



Yongrui Lin

SIG Instrumentation

For his diligent and thorough efforts in fixing critical bugs, improving documentation, and ensuring robust testing for SIG Instrumentation features

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The Last Releases



~~v1.31~~

OCI Image Volume Support
Device Health info in pod status
Kube-proxy nftables beta



v1.32

DRA Structured Parameter Support
Auto Remove support for PVC
Mutating admission policies using CEL



v1.33

Sidecar Container GA
In-place resource resize Beta
Node Topo label via downward API



v1.34

DRA core GA
KYAML alpha
Container restart rules
Discover cgroup driver from CRI
Load Env from file created in runtime

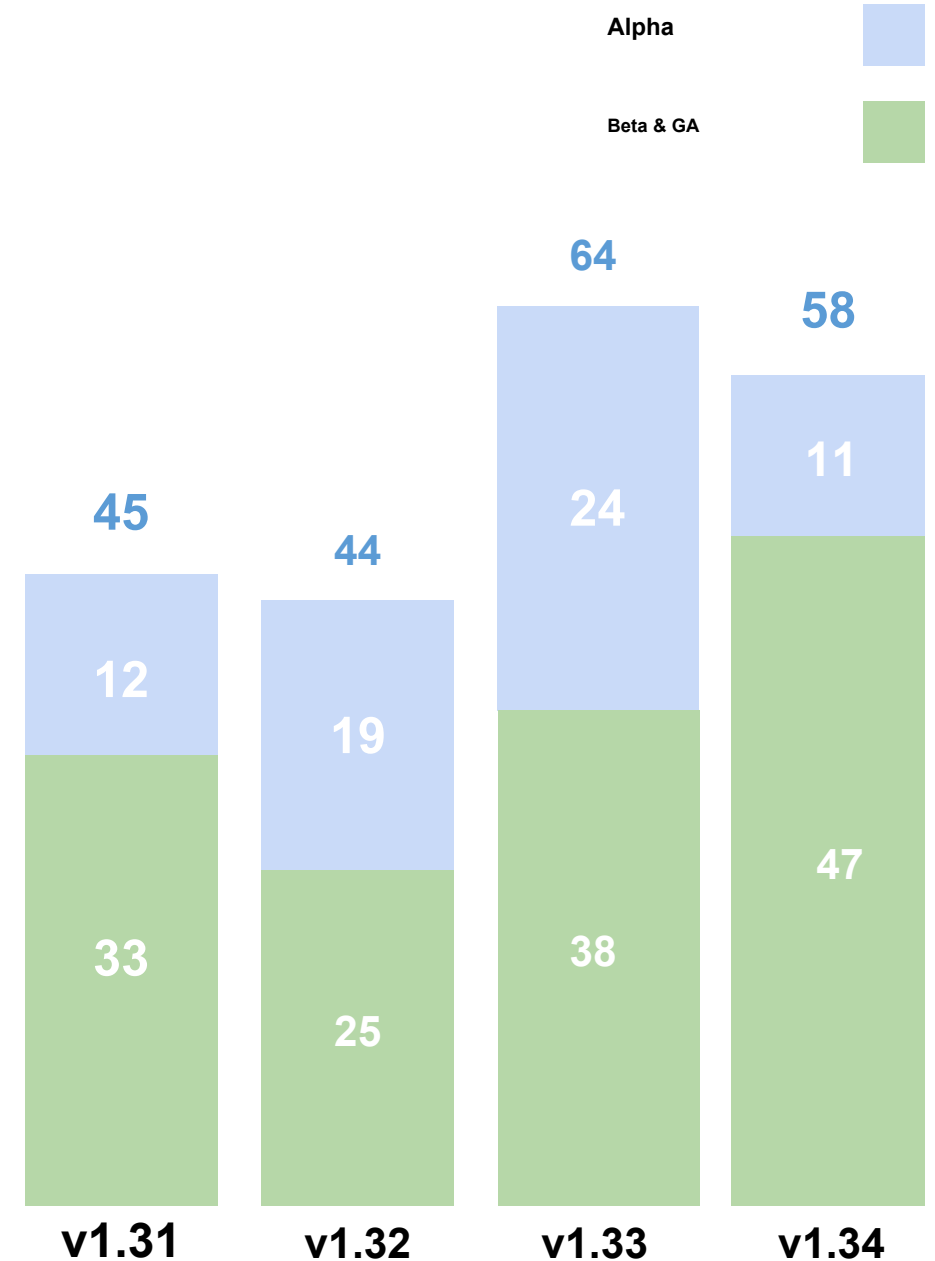
Trend analysis

Findings

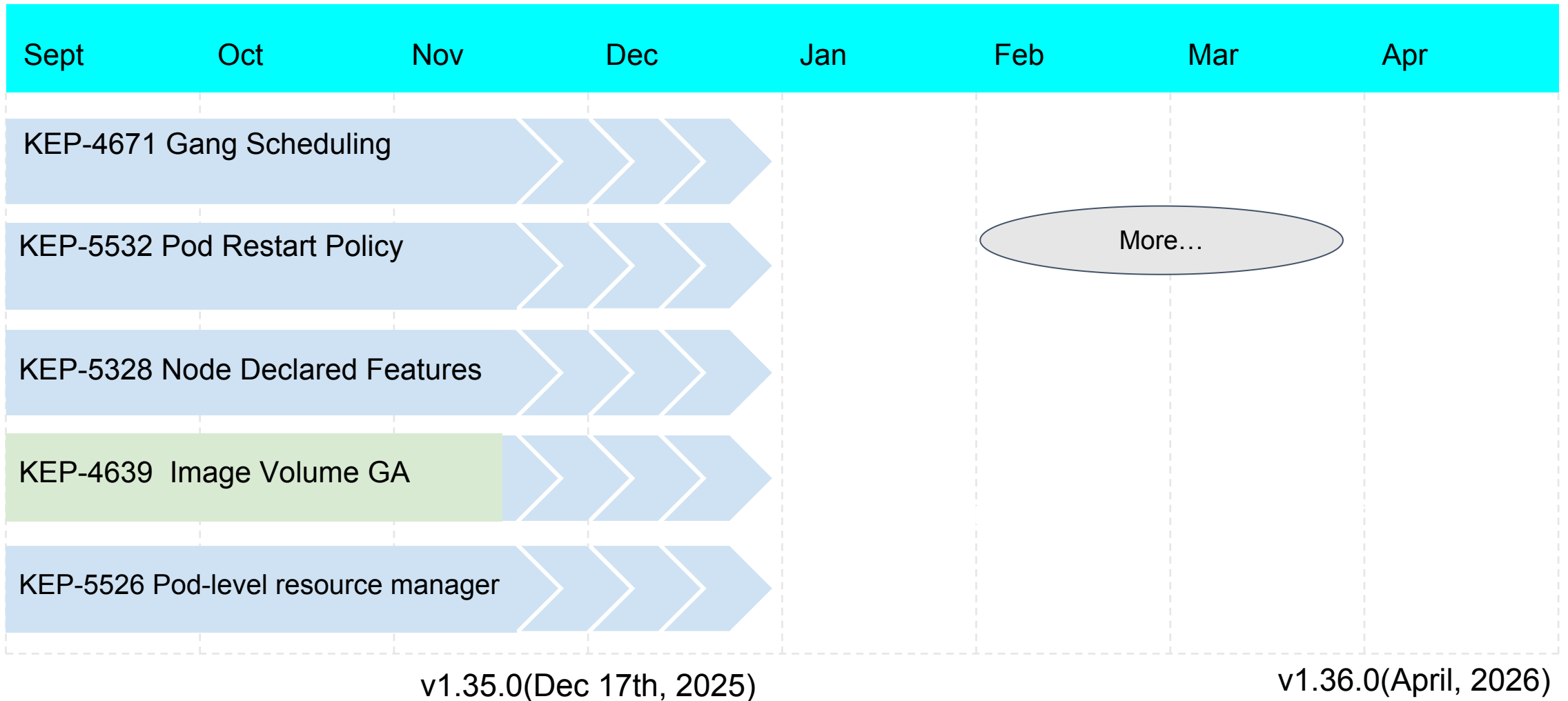
With more AI workloads on Kubernetes, there are more new KEPs.

Details:

- DRA divides into multi KEPs
- Observability & Security ↑
- Performance improvements for Large Cluster
- Safe Upgradation



Next releases: Timeline



Workgroups(AI related)

WG Batch

2022

Batch (eg. HPC, AI/ML, data analytics, CI) workloads.

- Batch API Group
- Multi Cluster support

Kueue, JobSet, Kjob



Kueue

WG Device Management

2024

DRA

- Extended Resource
- Claim Status
- Admin Access
- Device Taints and Tolerations
- Prioritized Alternatives

WG Serving

2024

Gather requirements for serving workloads.

- Inference Perf
- GAIE
- Service Catalog
- LWS



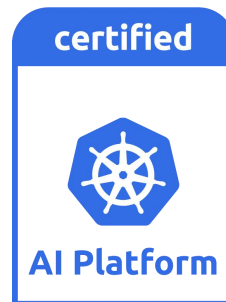
LWS

New Workgroups in 2025

WG AI Conformance

CNCF Kubernetes AI Conformance

- DRA
- Inference Gateway
- Gang scheduling
- Cluster Autoscaling
- Observability
- Security



WG AI Integration

integration of AI/ML control planes: AI agent (or multi-agent) systems to integrate with Kubernetes.

- MCP/A2A
- Agent Sandbox
- Agent Gateway etc.

WG AI Gateway

manage AI traffic

- Prompt Guards
- Token Rate Limiting
- Semantic Routing/Caching
- Response Risk
- Failure Modes
- Observability

[kube-agentic-networking](#)

New Workgroups in 2025

WG Node Lifecycle

- Declarative Node Maintenance **new!**
- Specialized Lifecycle Management(observability)**new!**
- Node declared features(1.35) **new!**
- Readiness gate **new!**
- Graceful node shutdown
- DRA device taints and tolerations

```
apiVersion: v1alpha1
kind: NodeMaintenance
metadata:
  name: "master-2-maintenance"
spec:
  node: "master-2"
  stage: Drain
  reason: "OS Upgrade"
  drainPlan:
    - podPriority: 5000
      podType: Default
    - podPriority: 15000
      podType: Default
    - podPriority: 3000
      podType: DaemonSet
```

WG Checkpoint Restore

checkpointing and restoring workloads (e.g., live migration, fault tolerance, debugging, snapshotting)

- GPU Checkpointing: CRIU
- Checkpoint and restore a container using kubectl
- Checkpoint and restore a Pod using kubectl
- Scheduling Integration

Deprecations & Community Support

Cgroup V1 Deprecation

Kubernetes 1.31: [Moving cgroup v1 Support into Maintenance Mode](#) (Aug 2024)

Cgroup V1 will be not supported by default in v1.35.

- FailCgroupV1 is by default true. (Dec 2025)

Cgroup v1 support will be removed **no earlier than v1.38** (end of 2026)

WG-LTS

A [draft planning](#) of `additional 12 months beyond the current support window of 14 months` for security fix.

[KEP-4330](#) Introduce compatibility version for Kubernetes control-plane upgrades

Navigating the Multi-Version Kubernetes Universe: How Emulation Version Shapes Your Contributions - Siyuan Zhang, Google

BUILDING B | LEVEL 2 | ROOM B206

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DRA reached GA in 1.34!

Dynamic Resource Allocation (DRA) is the new method for **requesting, configuring, sharing, and allocating resources** in Kubernetes, culminating a multi-year effort across SIGs.

DRA enhances Kubernetes' understanding of hardware, future-proofing it for AI/ML and other workloads

- Deep modeling of node-local resources (GPU, TPU, NIC, CPU, and more), including fractional devices.
- Modeling for multi-node resources (Compute Domain, Topology, TPU Slices).
- Flexible hardware modeling to support new technologies.
- Flexible resource requests for better resource obtainability.

The basics are GA, but there is still a lot of work underway - join us in contributing!



Learn more at the WG Device Management Talk!



DRA is GA! Kubernetes WG Device Management - GPUs, TPUs, NICs and More With DRA - Kevin Klues, NVIDIA & Patrick Ohly, Intel

📅 Tuesday November 11, 2025 4:15pm - 4:45pm

📍 Building C | Level 3 | Georgia Ballroom 3



Kevin Klues
NVIDIA
Distinguished Engineer



Patrick Ohly
Intel
Principal Engineer



John Belamaric
Google
Senior Staff Software Engineer

DRA Planning for 1.35, 1.36

KEP	Description	Owner SIG	Lead	Released			Targets			Notes
	See also the DRA enhancements project board .			1.32	1.33	1.34	1.35	1.36	1.37	
5004	DRA: Extended Resource Requests via DRA	Scheduling	yliaog			Alpha	<i>Beta</i>	?		Avoid user needing to know device plugin vs DRA
4817	DRA: Resource Claim Status	Node	LionelJouin	Alpha	Beta	Beta	<i>Stable</i>			Enables multi-network services
5018	DRA: Namespace Controlled Admin Access	Auth	ritazh		Alpha	Beta	<i>Beta</i>	<i>Stable</i>		No changes in 1.35, GA in 1.36
5055	DRA: Device Taints and Tolerations	Scheduling	pohly		Alpha	Alpha	<i>Alpha</i>	<i>Beta</i>		Health information, evict pods
4815	DRA: Partitionable Devices	Scheduling	mortent		Alpha	Alpha	<i>Alpha</i>	<i>Beta</i>		Underpins dynamic MIG/TPU, TPU slice allocation
4680	Resource Health Status in Pod Status	Node	jpsassine	Alpha	Alpha	Alpha	<i>Beta</i>	<i>Stable?</i>		Inform users about everything affecting their pods
5007	DRA: Device Binding Conditions	Scheduling	ttsuuubasa			Alpha	<i>Alpha</i>	<i>Beta</i>		Dynamically attach devices to nodes
5075	DRA: Consumable Capacity	Scheduling	sunya-ch			Alpha	<i>Alpha</i>	<i>Beta</i>		Basis for platform-mediated device sharing
4816	DRA: Prioritized Alternatives in Device Requests	Scheduling	mortent		Alpha	Beta	<i>Beta</i>	<i>Stable</i>		Second beta to add scoring
5491	DRA: List Types for Attributes	Scheduling	bg-chun, everpeace					<i>Alpha</i>		More flexible hardware topologies
5234	DRA: ResourceSlice Mixins	Scheduling	mortent					<i>Alpha</i>		Makes resource slices more compact
5194	DRA: ReservedFor Workloads	Scheduling	mortent					<i>Alpha</i>		Enable TPU slice scale > 256 nodes
5517	DRA: Native Resource Requests	Node	pravk03					<i>Alpha</i>		
5677	DRA: Resource Availability Visibility									UX improvement
5676	DRA: Explicit DeviceClass Fields	Scheduling								Performance optimization, especially for WAS
	Early Ideas									
	WAS integration	Scheduling								Probably part of the WAS KEP
	Likely something for contiguous group allocations	Scheduling	aws-ruhip							
	Likely something for a provisioning flow	Scheduling	johnbelamaric							
	Maybe connect consumable and partitionable	Scheduling	johnbelamaric							
	Maybe something for topological allocations	Scheduling	johnbelamaric							

DRA Planning for 1.35, 1.36



Kubernetes

CloudNativeCue

Since 2020

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4815	DRA: Partitionable Devices	Scheduling	morient		Alpha	Alpha	Alpha	Beta		Underpins dynamic MIG/TPU, TPU slice allocation
4480	Resource Health Status in Pod Status	Node	jsassine	Alpha	Alpha	Alpha	Beta	Stable	?	Inform users about everything affecting their pods
5002	DRA: Device Binding Conditions	Scheduling	tsuuubasa			Alpha	Alpha	Beta		Dynamically attach devices to nodes
5075	DRA: Consumable Capacity	Scheduling	sunya-ch			Alpha	Alpha	Beta		Basis for platform-mediated device sharing
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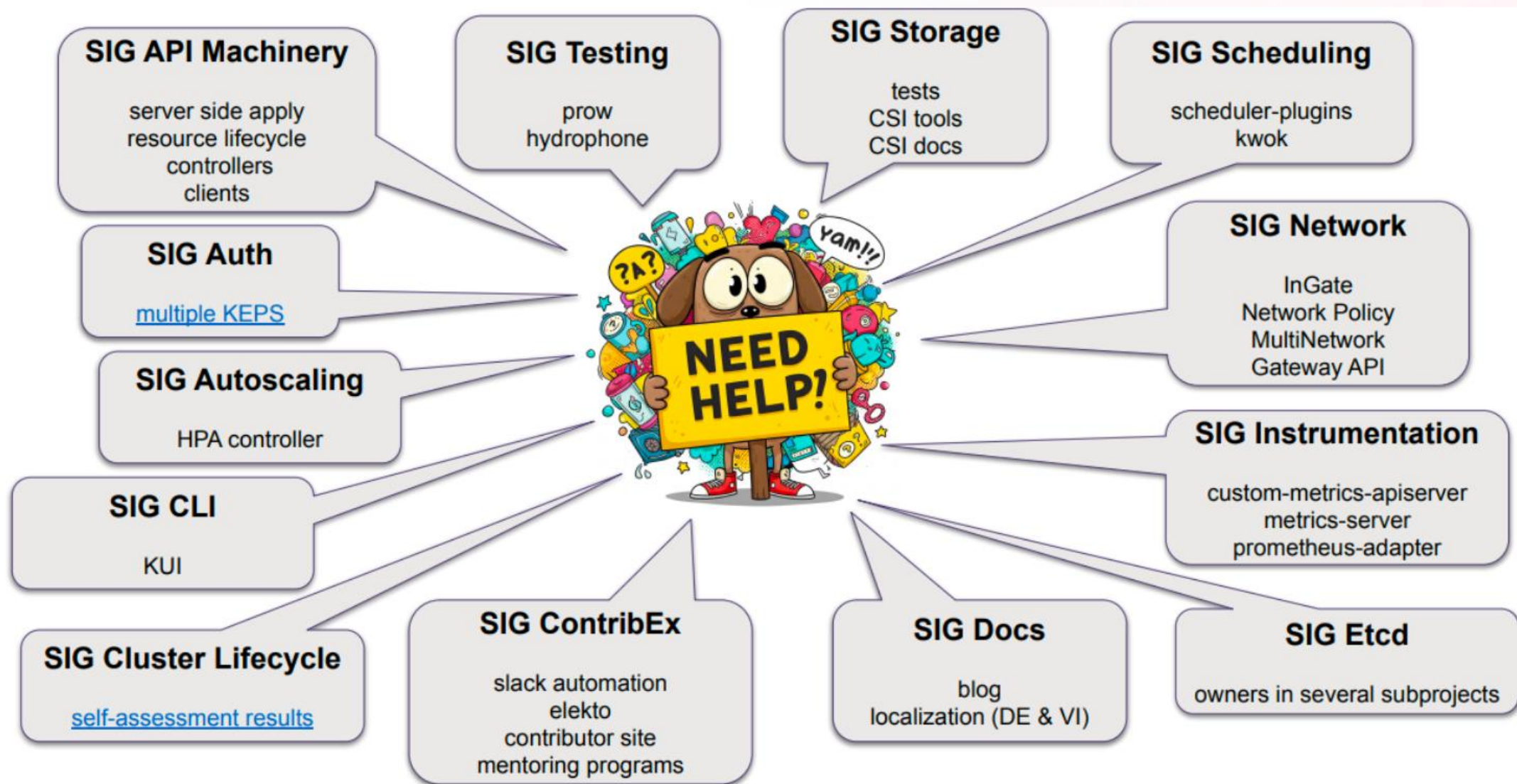
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Need Help



New Contributor Orientation

Kubernetes

New Contributor Orientation

July 2025



Phippy & Friends are Open Source (CC-BY) CNCF mascots. Learn more at: <https://phippy.io>

Welcome to the Kubernetes Contributor Community!

<https://www.kubernetes.dev/>

开放原子开源基金会企业AI创新实践赛

竞赛机制

■ 竞赛要求

本赛项面向企业与行业的特定垂类领域及应用场景，应利用企业人工智能开放平台(OPEA)、开源推理框架及服务架构（如PyTorch、OpenVINO、Ollama等）、结合主流开源大模型（如DeepSeek、千问等）以及其他开源AI技术、基于英特尔平台面向生产环境构建企业级AI解决方案。

作品形式主要为代码并包含安装流程、演示文档、演示视频等在内的其他项目资源，竞赛鼓励选手在初赛阶段使用相关硬件平台进行参赛项目的实际部署，并计划为进入决赛的团队提供相应的部署硬件环境。

参赛作品内容要求

初赛作品要求：

- (1) (必须) 项目技术报告：包含项目设计、技术架构、应用方向等整体概述，可以让审评官在短时间内了解项目/做品设计；
- (2) (必须) 项目代码：为原型或实际部署的代码，利用企业人工智能开放平台（OPEA）及其他推荐的开源服务及技术；
- (3) (必须) 视频演示：演示作品的运行效果，包含用户使用界面的基本操作及相应的输入输出信息在内的应用示例，后台模型的输入输出、性能及资源占用等关键技术点。
- (4) (推荐) 使用相应的硬件平台进行项目原型的部署，构建面向生产环境有相对综合成本优势的企业级AI解决及实施方案。
- (5) (可选) 其它展示作品的相关材料等；

决赛作品要求：

在初赛作品基础上，进一步完善作品的可部署性及可用性，并在相应的基于英特尔硬件平台中运行及展示



开放原子开源基金会
OPENATOM FOUNDATION



报名截止 12月14日